BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: PR - XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Min | **Maximum Marks:** 20

Note: [Urdu instructions printed at top of paper — best-effort] ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ جواب میں ایک سے زیادہ دائرے بھرے ہوں تو جواب غلط تصور ہوگا۔

Q1. Domain of function f(x) = |x| is _____.

(A) (0,∞) (B) (-∞,∞)

(C) (-∞,0) (D) (0,1)

Q2. The function f(x) = √(x-3) is continuous for all x and _____.

(A) x>3 (B) x≤3

(C) x<3 (D) x≥3

Q3. lim(x→0) (e^x - 1)/x = _____.

(A) 0 (B) e

(C) 1 (D) x

Q4. d/dx (cos x) = _____.

(A) sin x (B) -sin x

(C) tan x (D) sec x

Q5. If f(x) = $2x^3 - 3x^2 + 4$, then f'(2) = _____.

(A) 10 (B) 12

(C) 14 (D) 8

Q6. lim(x→0) tan x / x = _____.

(A) 0 (B) 2

(C) 1 (D) 3

Q7. If y = sin x then y''' = _____.

(A) sin x
(B) -sin x
(C) cos x
(D) -cos x

Q8. The graph of a function f(x) is concave downward on (a,b) where:

(A) f''(x)>0
(B) f''(x)<0
(C) f''(x)=0
(D) None of these

Q9. A function f(x) is decreasing if _____.

(A) f'(x)>0
(B) f'(x)<0
(C) f'(x)=0
(D) f''(x)=0

Q10. $\int_0^1 x^2\,dx$ = _____.

(A) 0
(B) -1/3
(C) 2/3
(D) 1/3

Q11. $\int \cot x\,dx$ = _____.

(A) ln|cos x|+c
(B) tan x+c
(C) ln|sin x|+c
(D) ln|tan x|+c

Q12. The slope of the line y-1=0 is _____.

(A) -1
(B) 0
(C) 1
(D) None of these

Q13. The distance between two points A(3,-2) and B(-1,-5) is _____.

(A) 10
(B) 11
(C) 5
(D) 4

Q14. The center of circle $x^2+y^2+2gx+2fy+c=0$ is _____.

(A) (g,f)
(B) (-g,-f)
(C) (g,-f)
(D) (-g,f)

Q15. The parabola $y^2=4px$, if p<0 then it is open _____.

(A) Down
(B) Up
(C) Left
(D) Right

Q16. If the eccentricity e<1, the conic is called _____.

(A) Ellipse
(B) Parabola
(C) Circle
(D) Hyperbola

Q17. $y^2/25 - x^2/4 = 1$ is an equation of _____.

(A) Circle
(B) Parabola
(C) Ellipse
(D) Hyperbola

Q18. Order of the differential equation $((d^3y/dx^3)^2 - (d^2y/dx^2)^4 + dy/dx + y - 0)$ is _____.

(A) 4
(B) 3
(C) 2
(D) 1

Q19. $f(x,y) = (\sqrt{x} + \sqrt{y})/(x+y)$ is a homogenous function of degree _____.

(A) -1/2
(B) 1/2
(C) 0
(D) 1

Q20. $\int_0^1$ e^(x^2) dx has _____.

(A) One actual solution
(B) No actual solution
(C) More than one actual solutions
(D) Both A and B

SECTION-B

Time Allowed: 2 Hours and 40 minutes

QQ2. Answer any TEN parts. Each part carries FIVE marks.

(1) Find the inverse function of $f(x) = (3x + 1)/5$.

(2) Evaluate $\lim(\theta \to 0) (\tan\theta - \sin\theta)/\sin^3\theta$

(3) Use first principal rule to determine the derivative of $f(x) = 5x^2 - 3x$?

(4) Find dy/dx by using any suitable rule of differentiation $y = x^2$e^(sin(x))

(5) If $f(x) = \ln(4x + 7)$, find f^iv (x).

(6) Evaluate $\lim(t \to 0)$ [sin(3t)/(3t) i + ln(2+t) j]

(7) Find the Maclaurin series expansion for the function $f(x) = \sin^2 x$.

(8) Evaluate $\int x \cos x \, dx$.

(9) Use the integration by parts to evaluate $\int x^2 \cos x \, dx$.

(10) Find the centroid of the triangle ABC whose vertices are A(3,5), B(4,6) and C(3,-1).

(11) Find the equation of parabola with vertex (-3,4) and passes through (5,11) and its line of symmetry is horizontal.

(12) If $f(x,y) = x^2y + xy^2$, then find fx and fy.

(13) Solve the initial value problem $dy/dx = 1/x^2$, $y(2) = 0$

SECTION-C

> Note: Attempt any THREE questions. All questions carry equal marks.

QQ3.

(a) Find dy/dx if $y = ((2x-5)/(x+4))^4$.

(b) Find the angle θ from the line $2x+4y-10=0$ to the line $5x-3y+1=0$

QQ4.

(a) Find the equation of tangent to the circle $x^2+y^2=9$ which makes an angle of 60° with x-axis?

(b) Find the critical value of $f(x) = x^2/(x-2)$

QQ5.

(a) Find the $\int(\pi/2 \text{ to } \pi) \cos(x/2 + \pi)\, dx$

(b) Find an equation of a circle which passes through the points (1,2), (3,-4) and (5,-6).

QQ6.

(a) Solve the differential equation $x\, dy/dx = x + y$

(b) Find the equation of hyperbola with vertices are at (5,0) and (-5,0) and one focus at (-7,0).